

We introduce the first hierarchical method for large and complex featured (hyper)graph generation in quasi-linear complexity

Motivation

- One-shot graph generators have quadratic complexity and poor scalability. In contrast, hierarchical approaches achieve quasi-linear complexity.
- Hypergraphs are more expressive than standard graphs and naturally model higher-order relationships, such as triangular faces in 3D meshes.
- Existing hierarchical generators for graphs and hypergraphs focus on topology generation, while node and edge features are ignored.
- Sequentially generating topology then features leads to poor results, highlighting the need for a joint topology–feature generation framework.

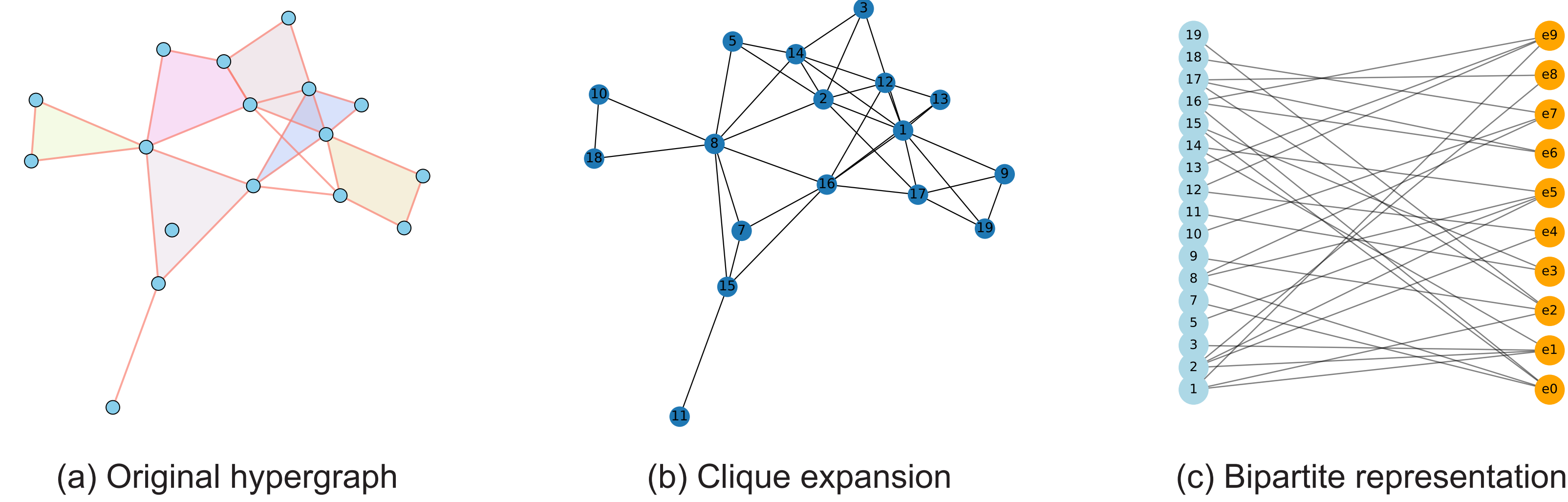
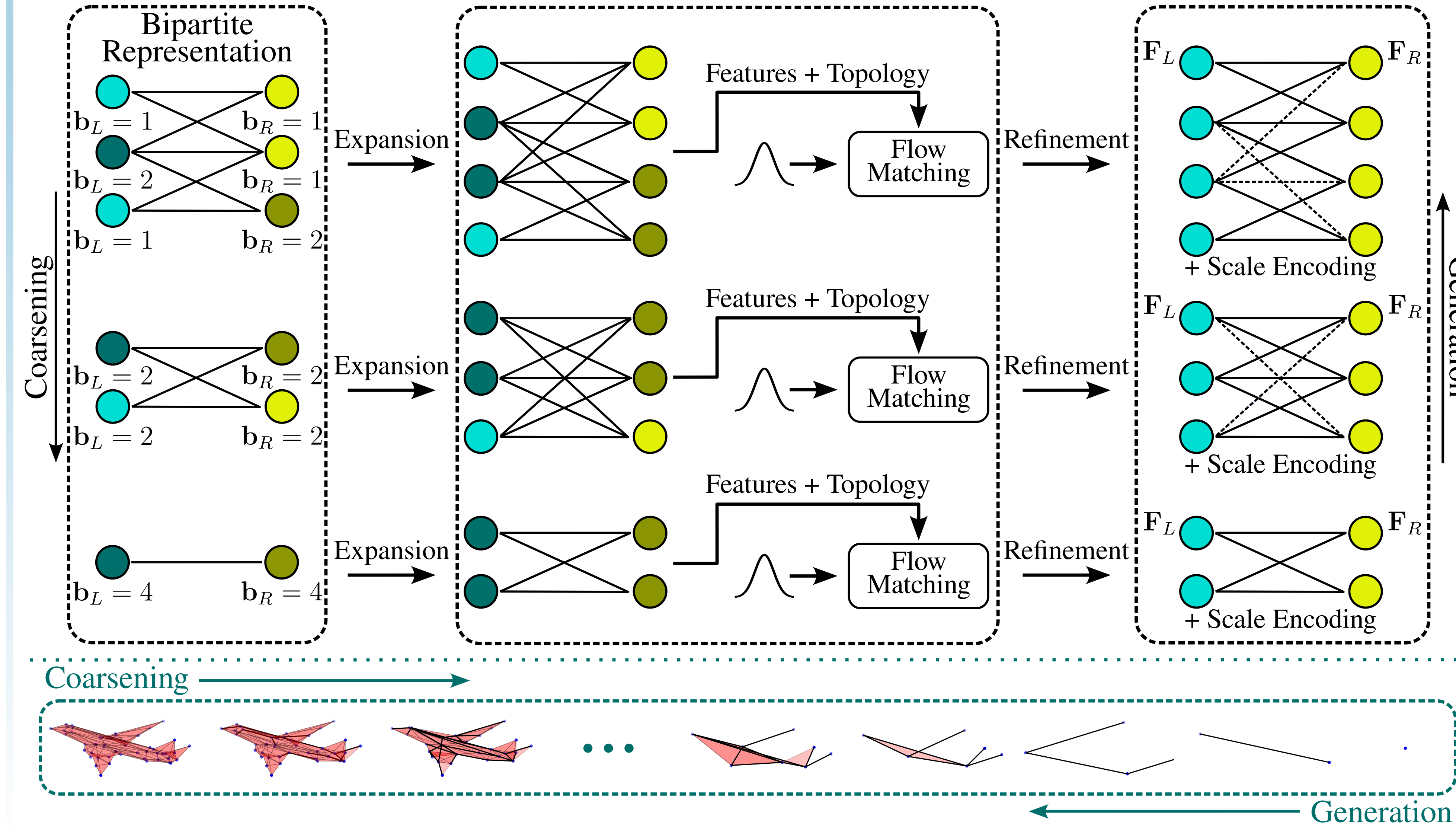


Figure. A hypergraph, its clique expansion and its bipartite representation

Pipeline



Ideas

We follow a coarsen-expand paradigm [1,2]: hypergraphs are progressively reduced [3], and a generative model learns to recover the missing structure and features. Our approach is based on three ideas:

- Node budgets:** Each cluster stores the number of nodes it represents, providing a notion of scale across different regions of the (hyper)graph.
- Coarsening:** Nodes are merged into coarse clusters while preserving the global structure; features are averaged, and budgets summed.
- Expansion:** The model reverses coarsening by jointly reconstructing connectivity, node features, edge features, and splitting budgets.

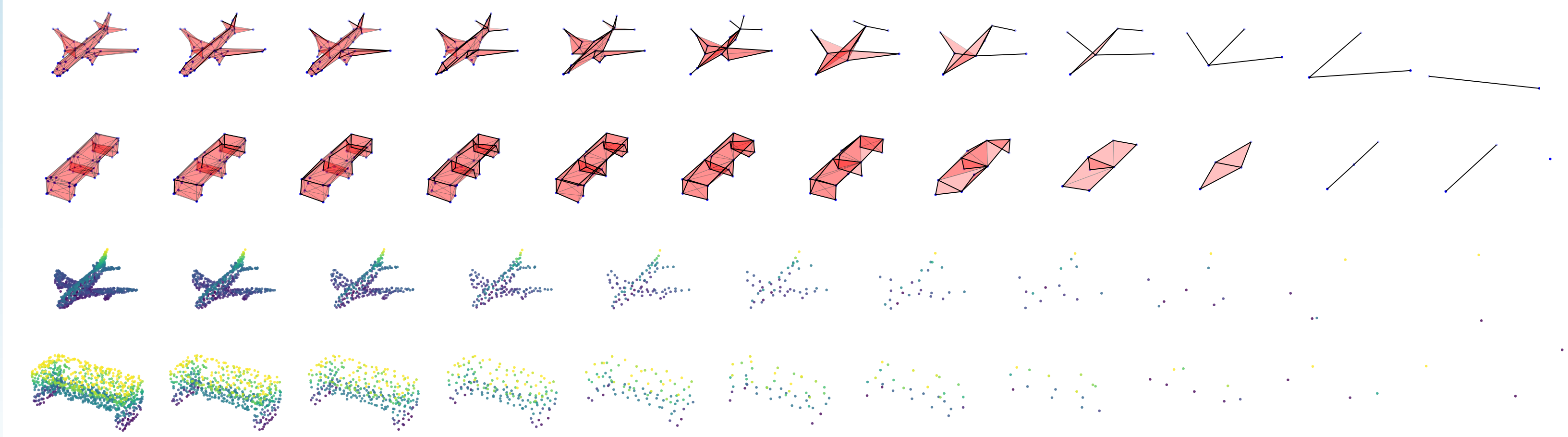
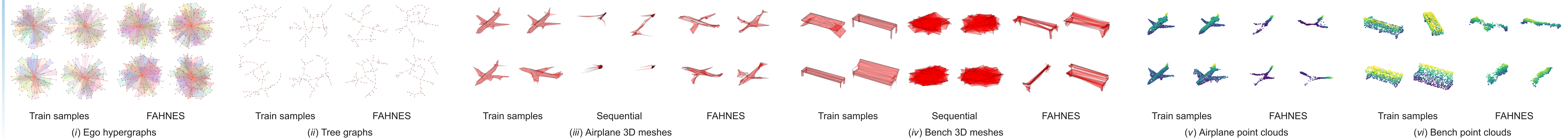


Figure. Examples of coarsening sequences for two 3D meshes and two point clouds.

Generated Examples



Numerical Results

Ego Hypergraphs ($n_{avg} = 109.71, std = 10.23$)			Tree Graphs ($n_{avg} = 64.0, std = 0$)			ManifoldNet40 Airplane ($n_{avg} = 51.86, std = 0.74$)			ManifoldNet40 Bench ($n_{avg} = 48.90, std = 3.52$)			Airplane Point Clouds ($n_{avg} = 971.14, std = 63.15$)			Bench Point Clouds ($n_{avg} = 987.31, std = 49.33$)		
Method	Valid (%) $\uparrow$	Spectral $\downarrow$	Method	Valid (%) $\uparrow$	Ratio $\downarrow$	Method	ChamDist $\downarrow$	ChamCov (%) $\uparrow$	ChamDiv $\uparrow$	Method	ChamDist $\downarrow$	ChamCov (%) $\uparrow$	ChamDiv $\uparrow$	Method	ChamDist $\downarrow$	ChamCov (%) $\uparrow$	ChamDiv $\uparrow$
HyperPA	0.0	0.237	HSpectre	100.0	4.0	Sequential	0.082 ± 0.074	34.8 ± 0.0	0.129 ± 0.011	Flow Matching	0.169 ± 0.007	13.3 ± 0.0	0.157 ± 0.003	Flow Matching	OOM	OOM	OOM
VAE	0.0	0.133	BwR	0.0	11.4	WGAN	0.124 ± 0.021	10.4 ± 3.5	0.037 ± 0.016	DiGress	0.128 ± 0.015	17.3 ± 3.3	0.063 ± 0.099	DiGress	OOM	OOM	OOM
GAN	0.0	0.230	DiGress	90.0	1.6	Flow Matching	0.085 ± 0.014	20.9 ± 10.0	0.063 ± 0.005	DeFoG	0.108 ± 0.019	30.7 ± 2.5	0.112 ± 0.007	DeFoG	OOM	OOM	OOM
Diffusion	0.0	0.190	DeFoG	96.5 ± 2.6	1.6 ± 0.4	VAE	0.231 ± 0.208	26.9 ± 11.7	0.191 ± 0.130	FAHNES	0.280 ± 0.223	27.5 ± 9.9	0.195 ± 0.107	FAHNES	0.094 ± 0.006	22.2 ± 14.5	0.085 ± 0.026
HYGENE	<u>90.0</u>	0.004	FAHNES	100.0 $\pm$ 0.0	<u>1.8 ± 0.8</u>	FAHNES	0.048 ± 0.003	<u>31.9 ± 13.2</u>	0.101 ± 0.011	FAHNES	0.064 ± 0.005	42.2 ± 13.9	0.243 ± 0.037	FAHNES	0.130 ± 0.001	15.8 ± 10.6	0.191 ± 0.114

References:

- [1] Andreas Bergmeister, Karolis Martinkus, Nathanaël Perraudin and Roger Wattenhofer, *Efficient and scalable graph generation through iterative local expansion*, ICLR 2024.
 [2] Sucheng Ren, Qihang Yu, Ju He, Xiaohui Shen, Alan Yuille and Liang-Chieh Chen, *FlowAR: Scale-wise autoregressive image generation meets flow matching*, ICML 2025.

- [3] Andreas Loukas, *Graph reduction with spectral and cut guarantees*, JMLR, 2019.

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Conclusions

- We propose a novel hierarchical approach for the generation of featured graphs and hypergraphs, bridging ideas from coarse-to-fine graph generation [1] and hierarchical image generation [2].
- This marks an important milestone toward scalable hierarchical generative models for graphs and related objects.

Next: Improve the robustness of the generation process.